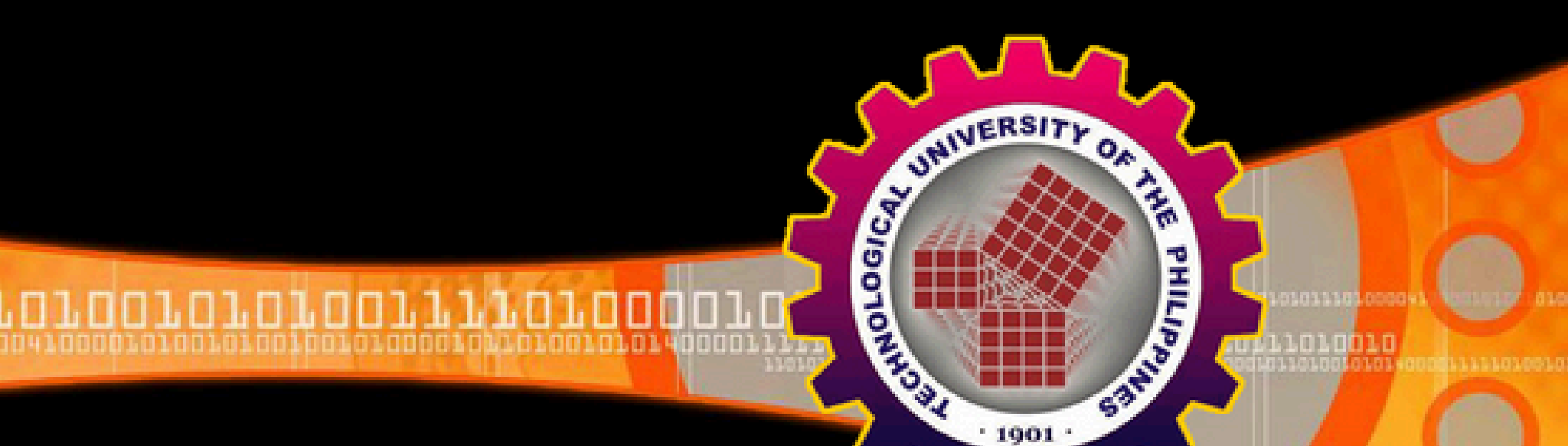


Automated Duck Egg Sorter

Bachelor of Science in Information Technology
Technological University of the Philippines - Taguig Campus

Members



BSIT-S-2A-T | Group 2

- Pasteteo, Juliane Caye
- Jumoc, Ernz Rabbi
- Ofracio, Kevin

Description

The Automated Duck Egg Sorting System aims to make egg sorting faster and more accurate. Sorting eggs manually takes time and can lead to mistakes. This system uses a camera, color sensor, candling light, and weight scale to classify eggs into three types: red eggs (salted eggs), Balut, and Penoy. A conveyor belt moves the eggs, while servo motors adjust a flat surface divider to guide them into their respective sections. An OLED Screen displays the sorting results, making the process efficient and easy to monitor.

Main Components and its Function

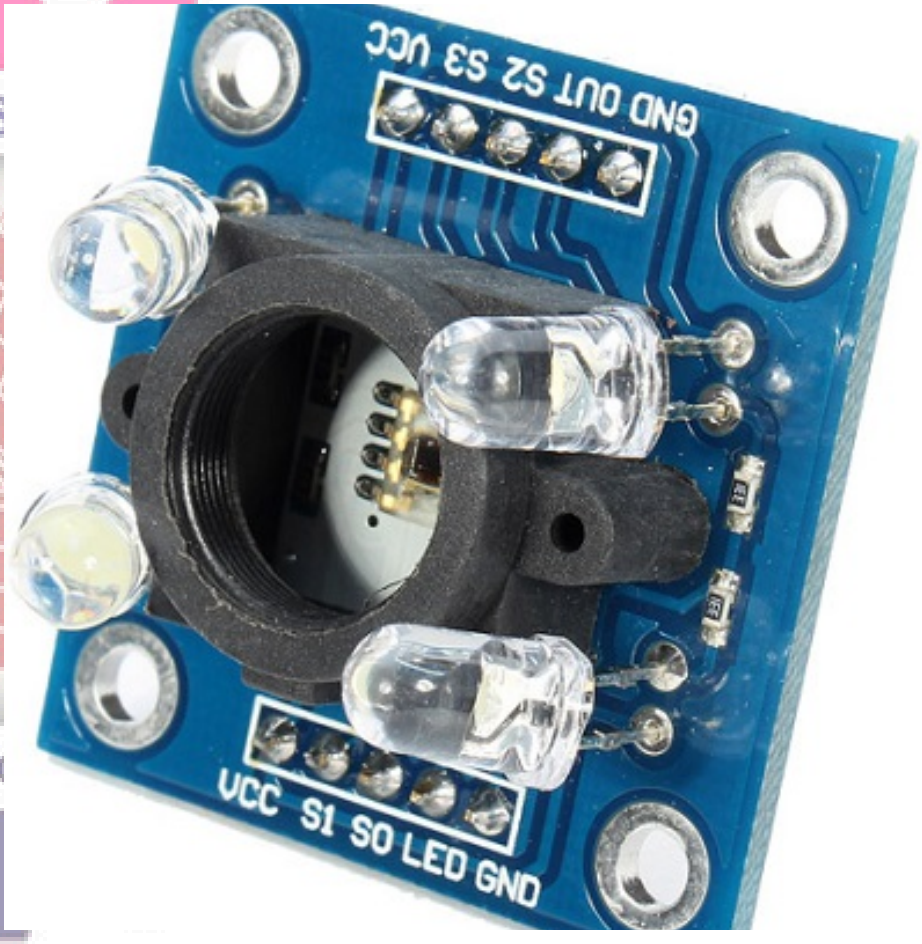
- **ESP32 Wroom-32**
  - The microcontroller that serves as the brain of the system, processing sensor data and controlling various components.



Main Components and its Function

Color Sensor TCS3200

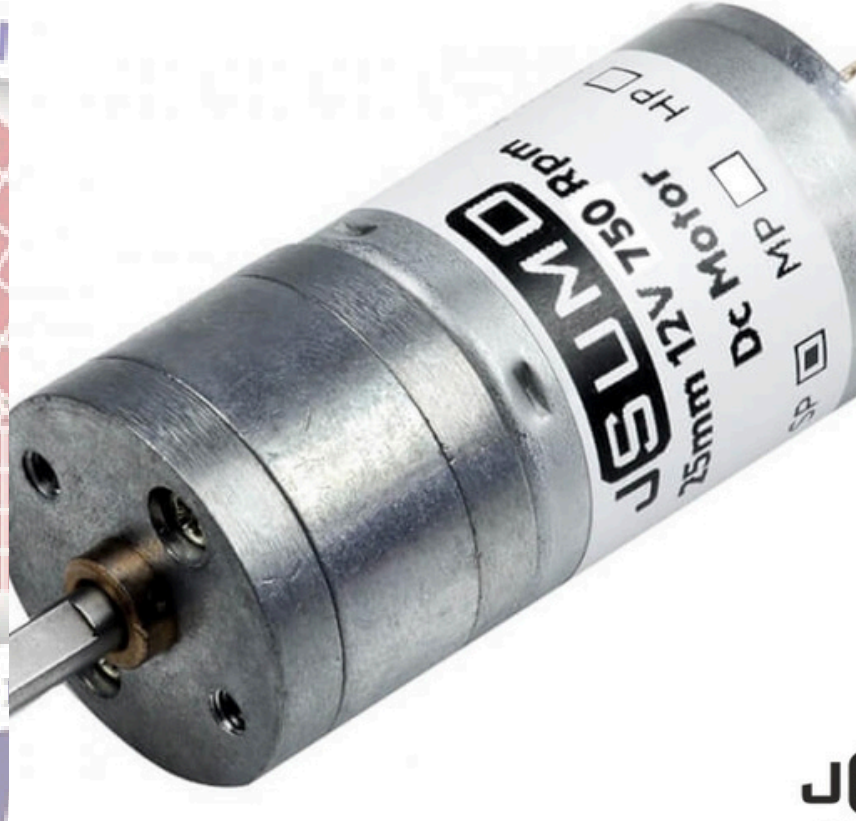
- The TCS3200 programmable color light-to-frequency converter combines configurable silicon photodiodes and a current-to-frequency converter on a single monolithic CMOS integrated circuit. It is used to separate red eggs and normal color eggs.



Main Components and its Function

DC Motor

- A DC motor is used to run the conveyor belt, pulling the rubber belt and rotating at an interval.

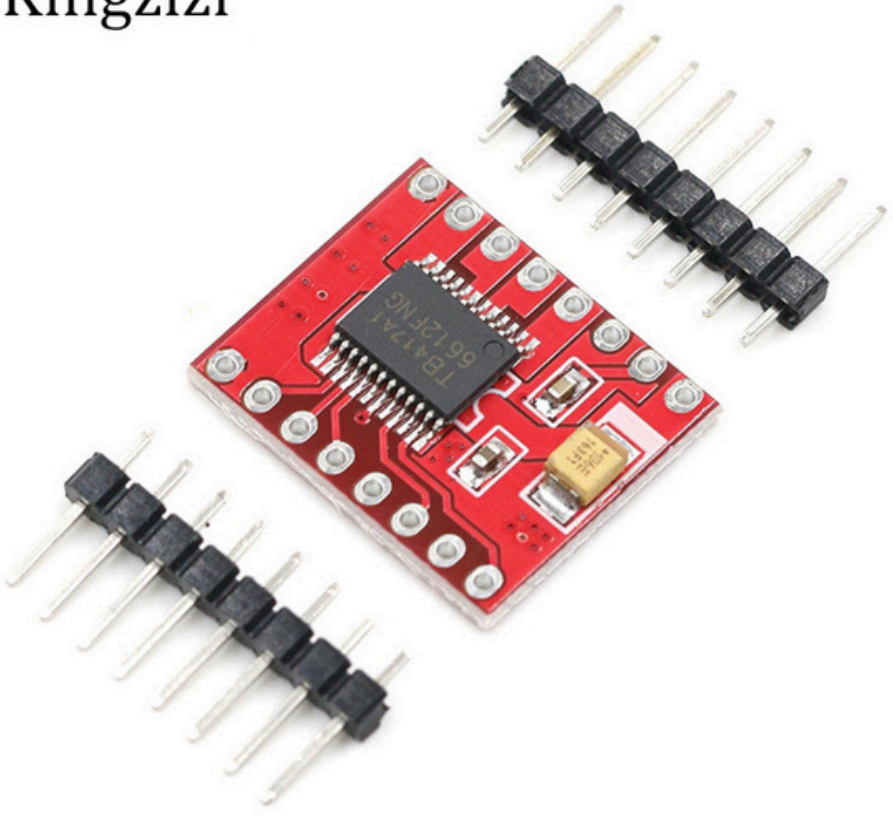


Other Components and its Function

Motor Driver

- The TB6612FNG motor driver can control up to two DC motors at a constant current of 1.2A (3.2A peak). It is used to configure the DC motor.

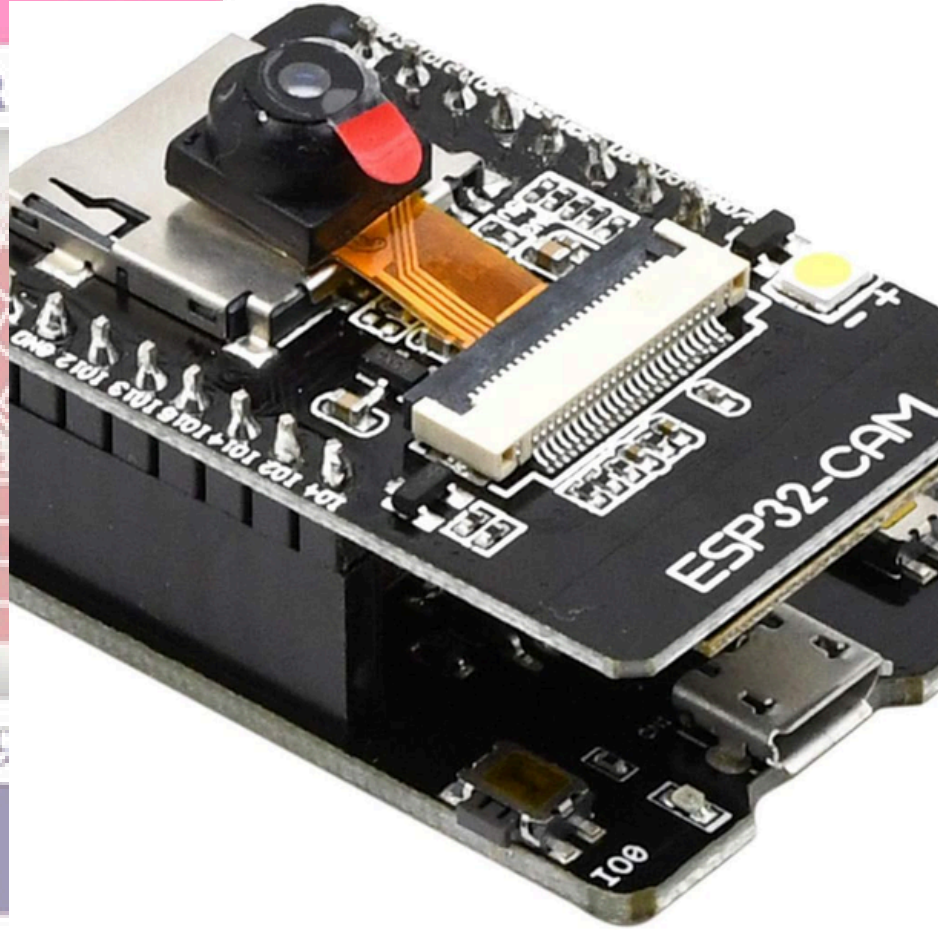
Kingzizi



Other Components and its Function

Camera Module

- The ESP32-CAM is a small size, low power consumption camera module based on ESP32. It comes with an OV2640 camera and provides onboard TF card slot. It is used to separate the balut and penoys.



Other Components and its Function

Candling Light

- Illuminates the egg to check for embryo presence.

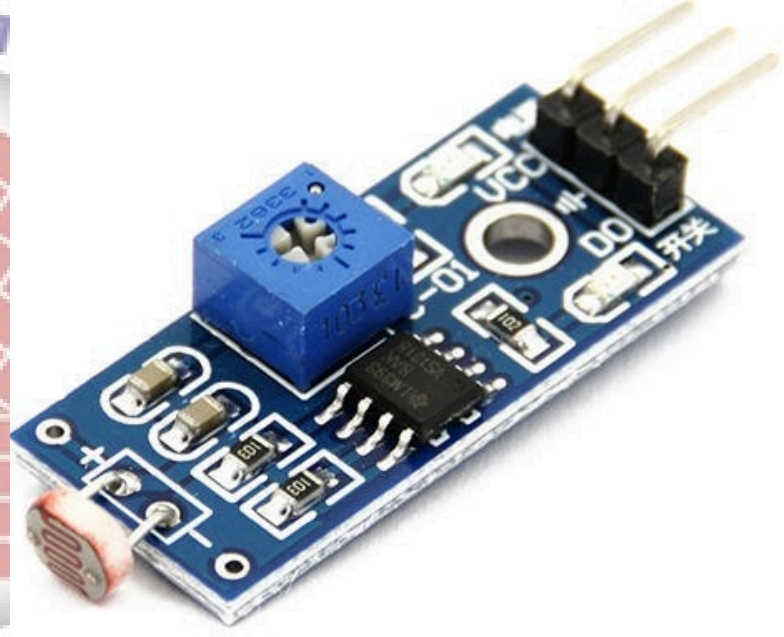




Other Components and its Function

Light Sensor

- Illuminates the egg to check for embryo presence.



Other Components and its Function

Servo Motor

- A servo motor, or "servo," is a specialized motor that provides precise control of angular or linear position, velocity, and acceleration, often used in robotics, automation, and radio-controlled applications. Moves eggs through the sorting process.



Other Components and its Function

Conveyor Belt

- Moves eggs through the sorting process.



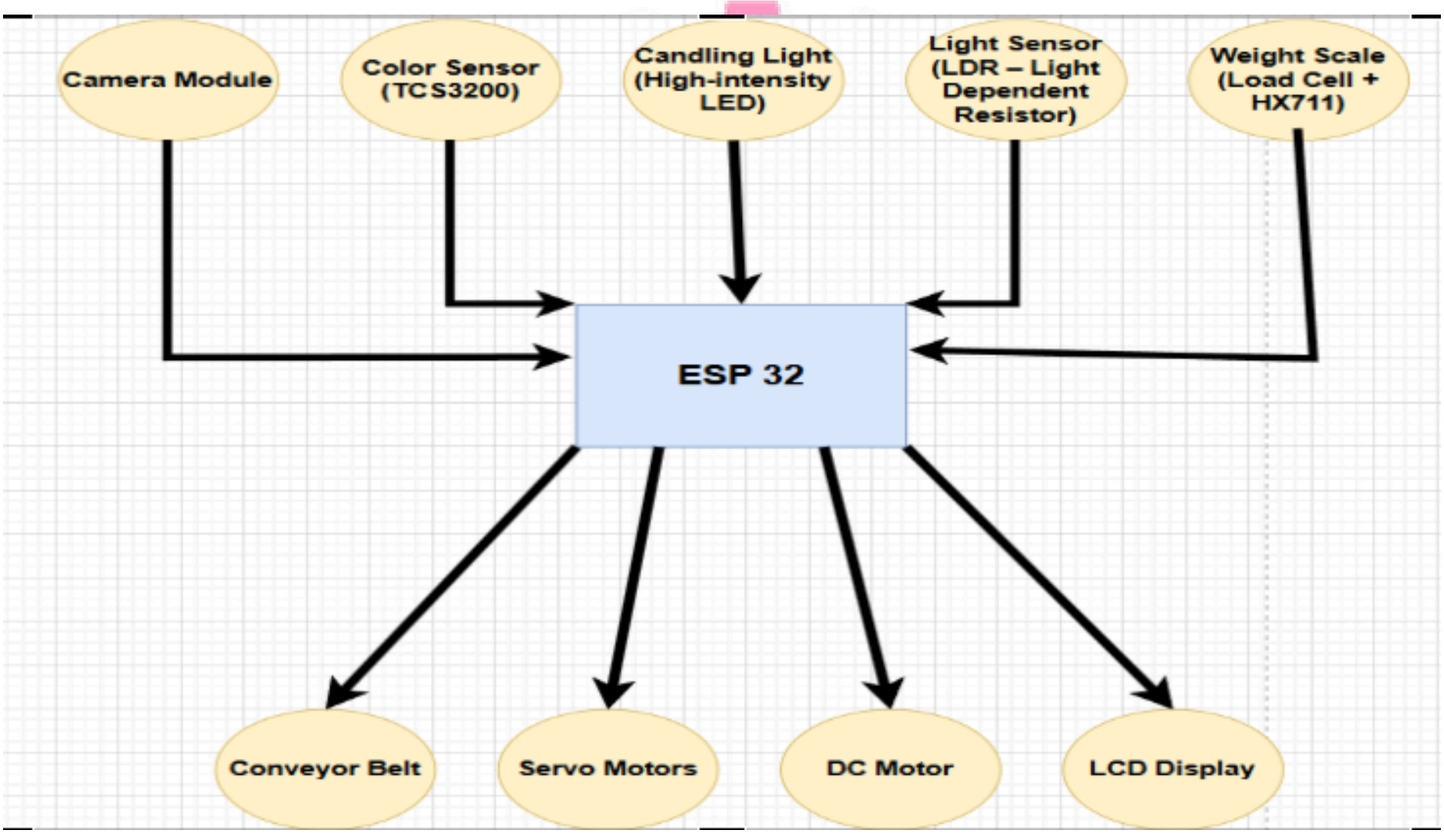
Other Components and its Function

I2C OLed

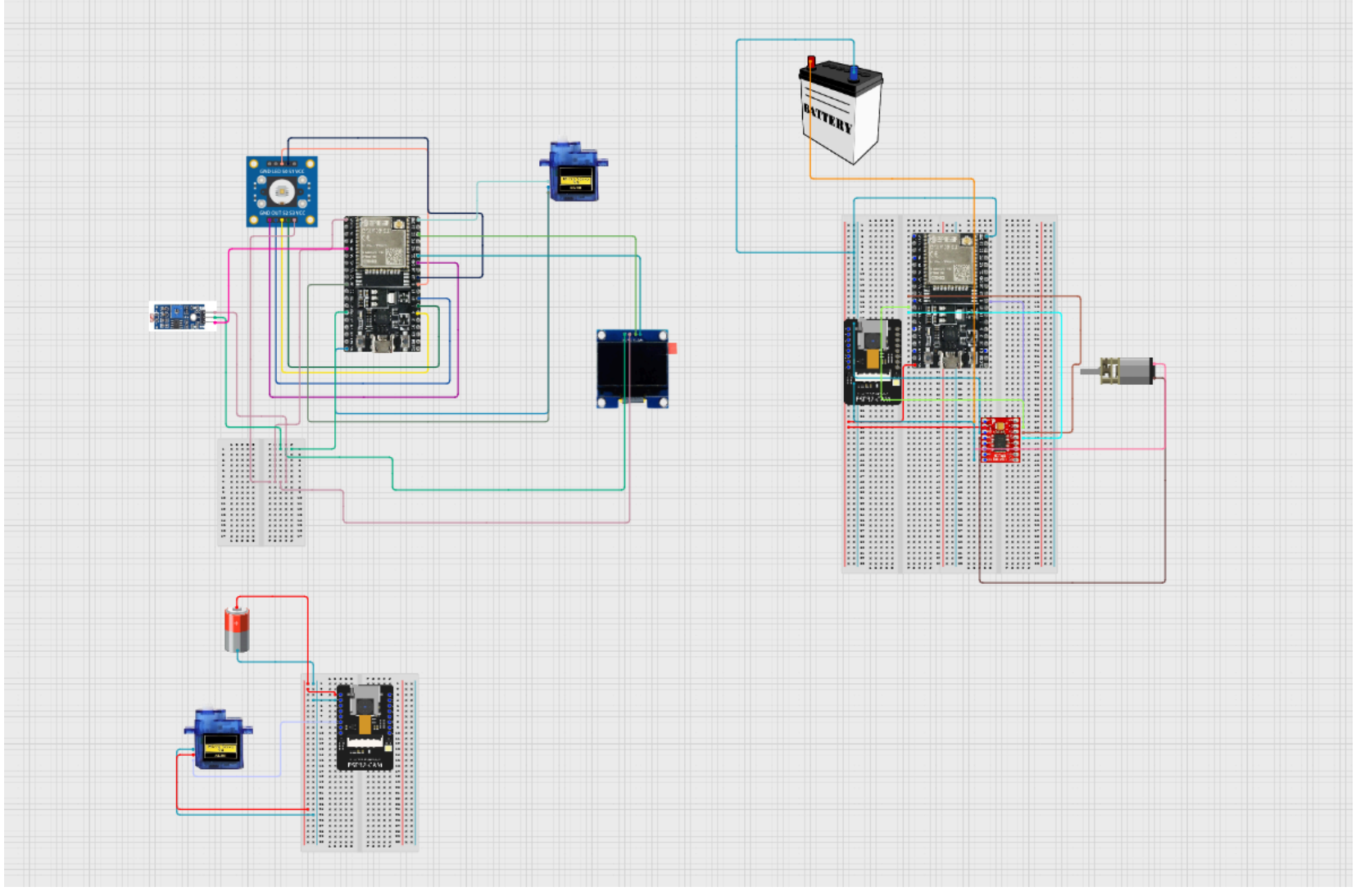
- OLED (Organic Light Emitting Diode) and LCD (Liquid Crystal Display) are two main types of display technologies, with OLED offering superior contrast, color accuracy, and viewing angles due to its self-emissive pixels, while LCDs use a backlight and are generally more affordable. Shows egg classification results in real-time.



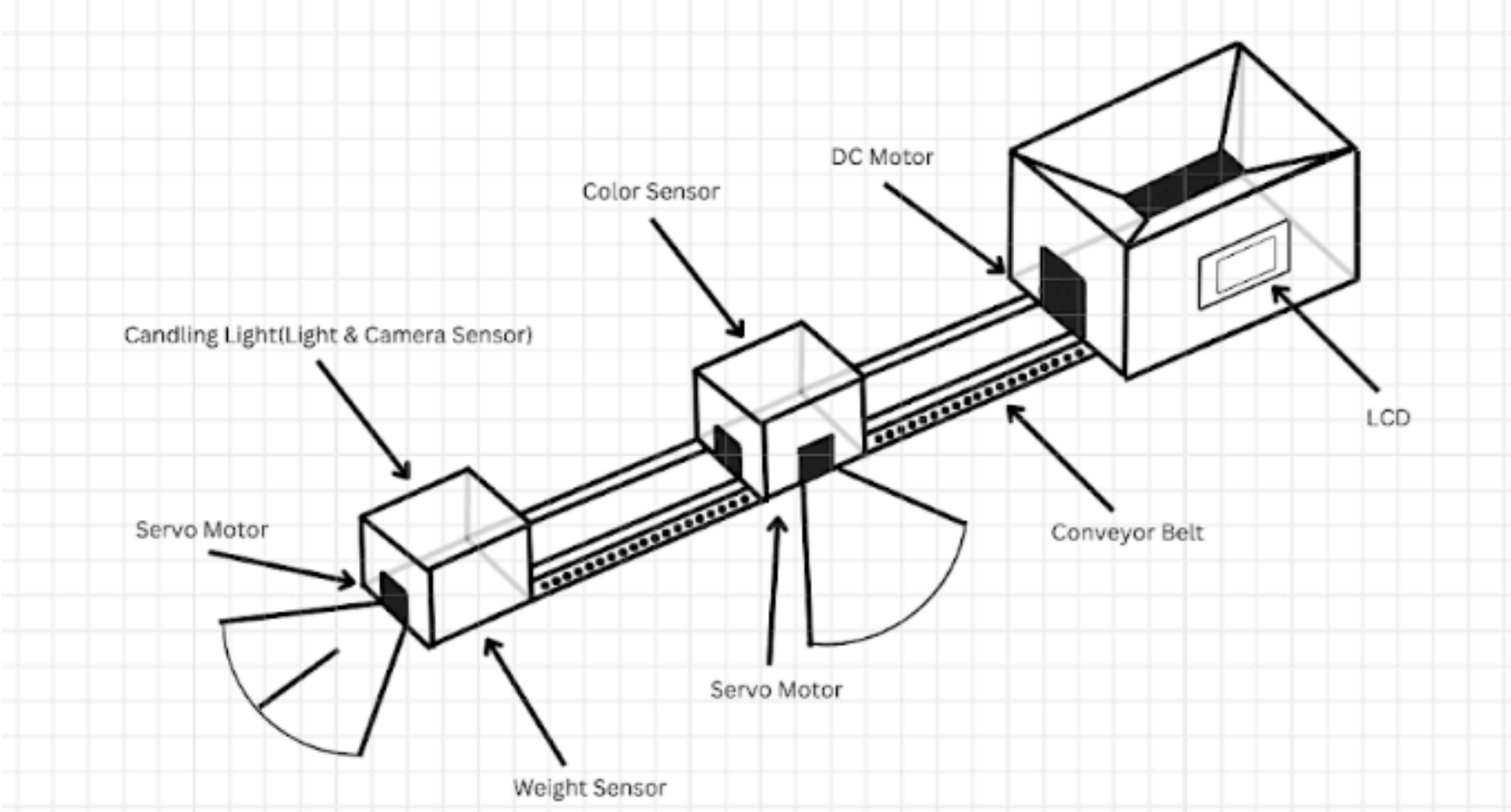
Block Diagram



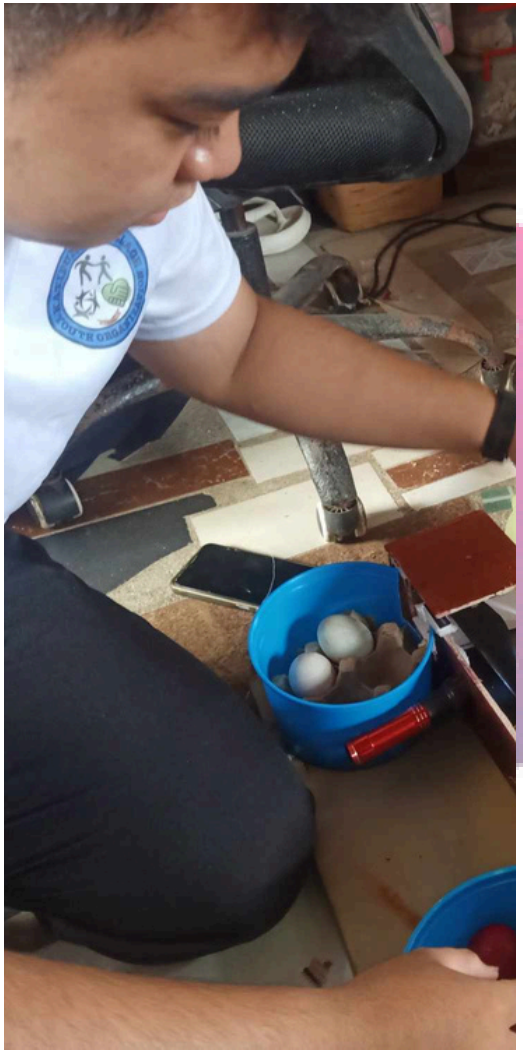
Schematic Diagram

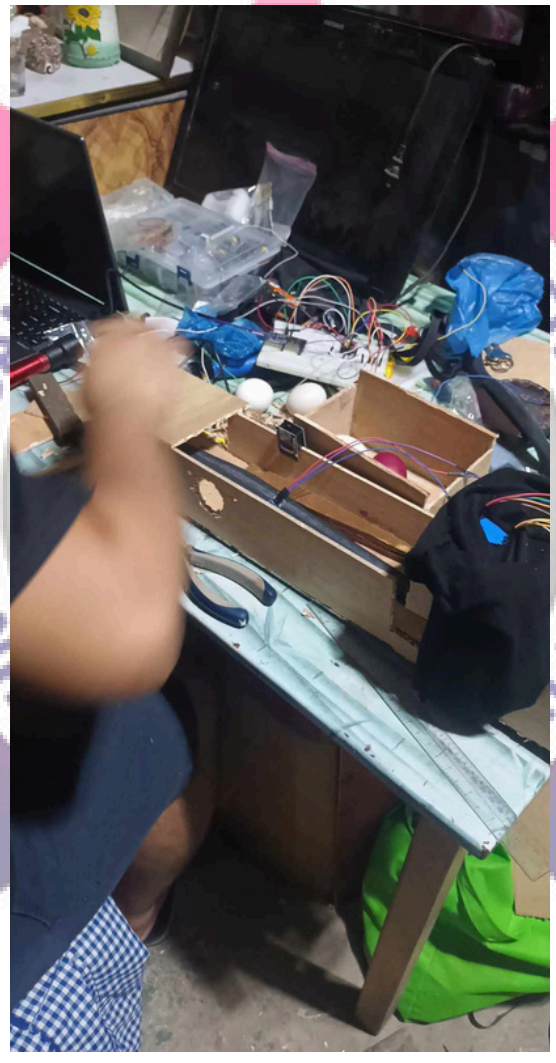


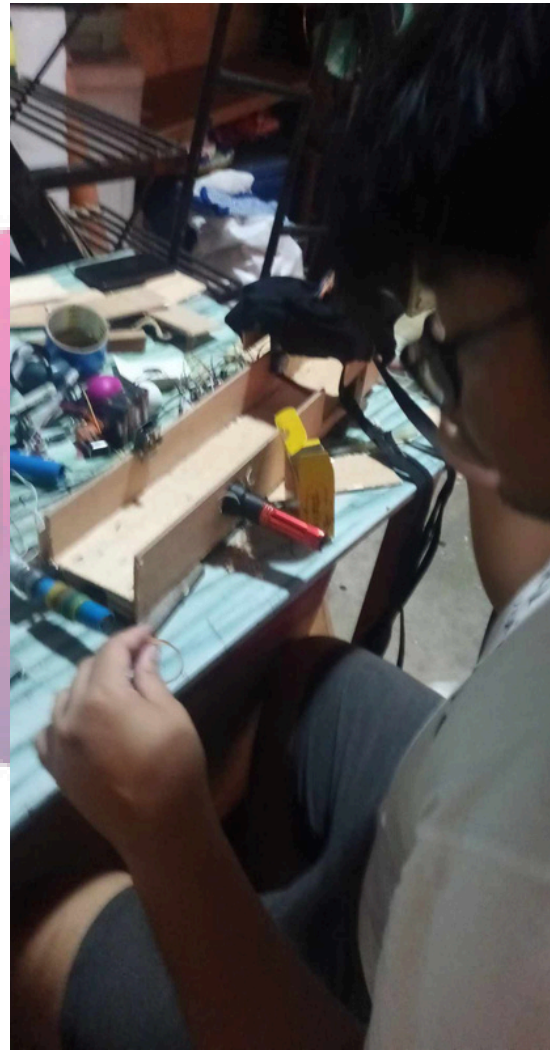
Prototype


DC Motor
Color Sensor
Candling Light(Light & Camera Sensor)
LCD
Servo Motor
Conveyor Belt
Servo Motor
Weight Sensor

Process

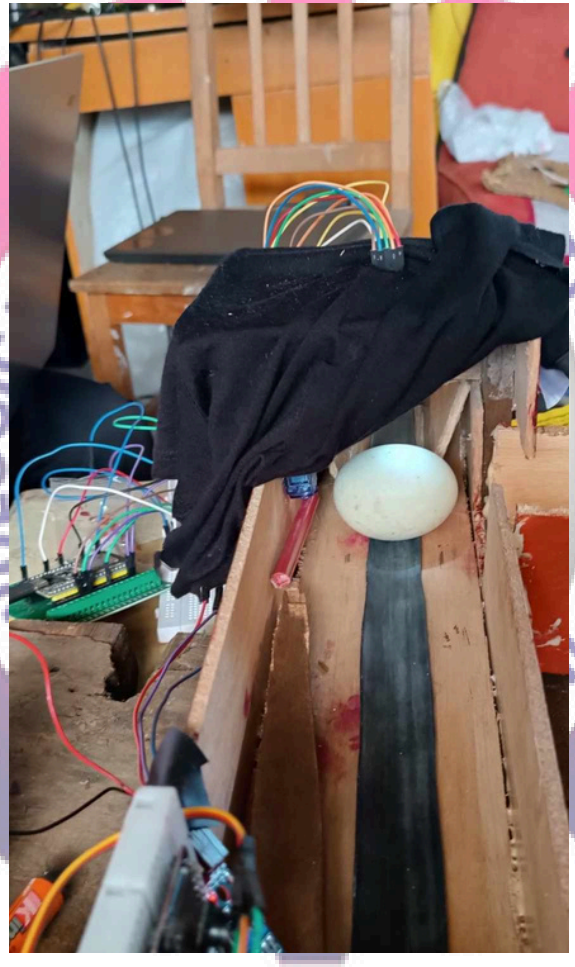






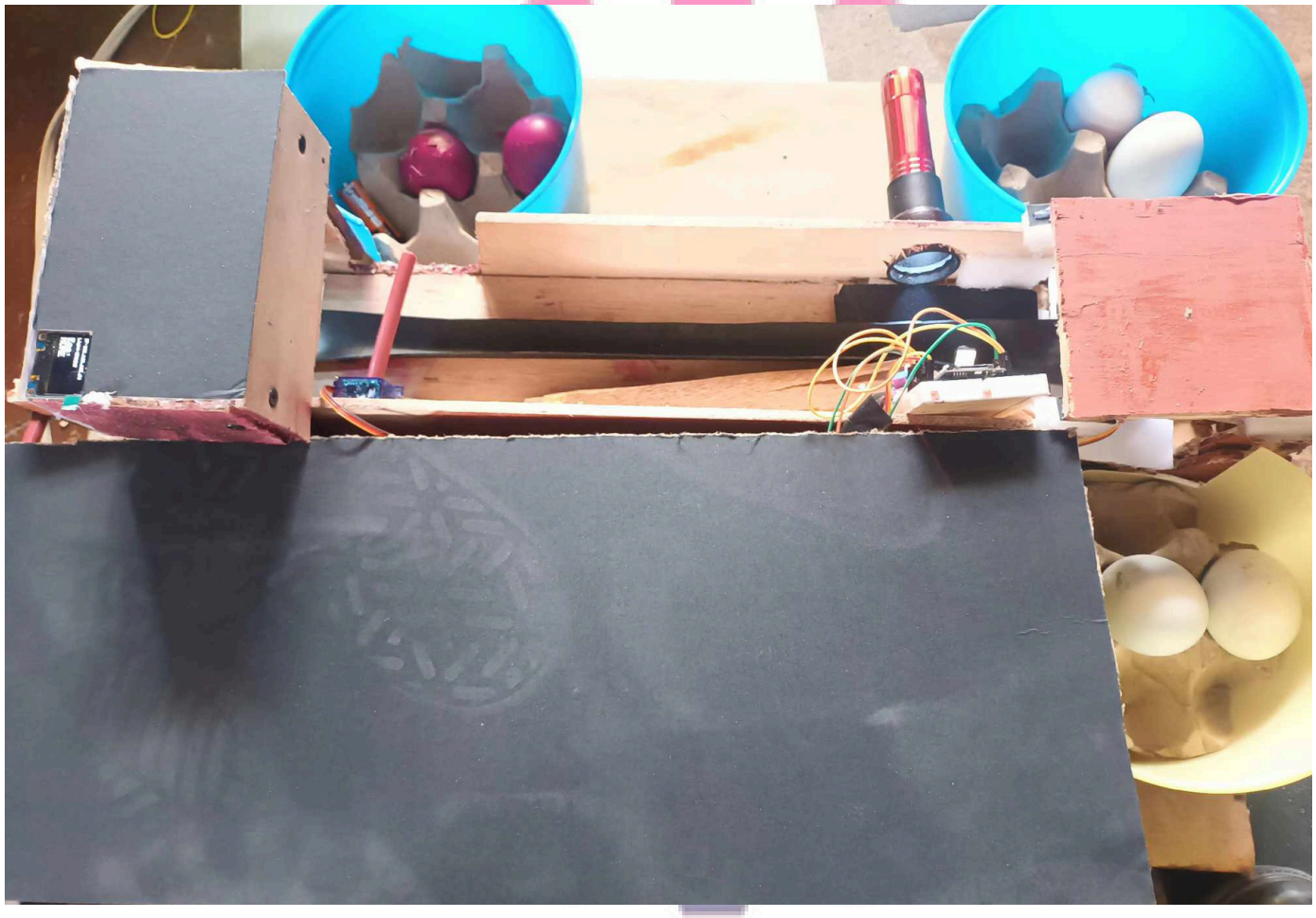
Process







Final Output



References

- *Duck Egg Setter System with Sorting Process Using Image Processing.* (2023, November 19).
- IEEE Conference Publication | IEEE Xplore. https://ieeexplore.ieee.org/document/10589163
- Magwili, G. (2024). An automated candling system for duck egg fertility detection, sorting,
- and counting via digital image processing. https://www.academia.edu/95457508